

SBIR Phase III

PERFORMANCE WORK STATEMENT (PWS)

for

SaaS Supply chain AI (S2A)

for

AI Analyses

of

Reliability, Logistics, and Sustainment Processes

of

Line Replaceable Units (LRU)s

SBIR Topic # AF181-038

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Acronym List

ADVANA	DoD platform that serves as an enterprise data and analytics environment
AF	Air Force
AFB	Air Force Base
AFSC	Air Force Sustainment Center
AI	Artificial Intelligence
ATO	Authority To Operate
App	Software Application
BIT	Built In Test
BLADE	Basing and Logistics Analytics Data Environment
CBM+	Condition Based Maintenance Plus
cATO	continuous Authority To Operate
CDRL	Contract Data Requirements List
CoA	Course of Action
COTR	Contracting Officer's Technical Representative
DD	Department of Defense
DID	Data Item Description
GCSS-AF	Global Combat Support System – Air Force
KPI	Key Performance Indicator
LRU	Line Replaceable Unit
MDS	Mission Design Series
MICAP	Mission Impaired Capability Awaiting Parts
NIIN	National Item Identification Number
NRTS	Not Repairable at This Station
OO-ALC	Ogden Air Logistics Center
P1	DoD PlatformOne
PWS	Performance Work Statement
RAMC	Reliability, Availability, Maintainability, and Cost
RDT	Reliability Digital Twin
RLDT	Reverse Logistics Digital Twin
RTS	Repairable at This Station
SaaS	Software as a Service
SBIR	Small Business Innovation Research
SBSS	Standard Base Supply System
SCAI	Supply Chain AI
SCAIT	Supply Chain AI Tool
SCM	Supply Chain Manager
SCMS	Supply Chain Management Squadron
SCMW	Supply Chain Management Wing
SLA	Service Level Agreement
SLO	Service Level Objective
SME	Subject Matter Expert
WUC	Work Unit Code

1.0 Objective

The objective of this Phase III Small Business Innovation Research (SBIR) task is to support the 448th SCMW by generating monthly AI analyses of reliability, logistics, and sustainment processes of Line Replaceable Units (LRU)s. The analysis results will be available as reports and also accessible through SaaS cloud software dashboards.

This task will rely on Supply Chain AI Technology (SCAIT) for Sustainment Data Analysis and Reporting. This Phase III will apply the SCAIT tools to a larger cross-section of sustainment processes managed by the 448th SCMW and make the tools available as SaaS software service in a Pilot Project. Data for selected LRUs managed by the 448th SCMW across sustainment operations at Tinker AFB, Hill AFB, and Robins AFB will be analyzed each month through the performance period. The analysis results will provide positive operational impact from the deployment of the SBIR technology.

This effort will focus on scaling the processes, analytics, reporting tools, and software from the previous SBIR Phase III contracts FA8222-21-C-3001, FA8212-22-C-0014, and FA8212-24-C-0005. This project will deliver AI Analytics Results for many selected LRUs to many users as SaaS Software Application.

The initial SCAIT tools were developed in AF141-206 SBIR topic Phase II and Sequential Phase II projects managed at Hill AFB. The tools were matured, and additional functions developed, within AF181-038 topic Phase II project managed at Edwards AFB. The AI contracts included SBIR Phase I FA8222-14-M-0012, Phase II FA8222-16-C-0001, SBIR Sequential Phase II FA8222-19-C-0019, Phase III FA-8222-18-C-0015 as well as SBIR Phase I FA9300-18-P-1010 and Phase II FA9300-20-C-2004.

2.0 Background

This SBIR Phase III targets efficiency improvements of equipment maintenance and logistics support processes in the Air Force Supply Chain. The sustainment processes can be viewed as reverse logistics system that includes repairs, testing, procurement, and retirement of the parts. The system is driven by part failures, removals, and upgrades. At present, Air Force collects sustainment data, but analytical processing of the data is not well established.

DoDD 5000.2 and DoDI 5000.91 provide high-level requirements to Systems Lifecycle Management. The DoD requires use of Reliability, Availability, Maintainability, and Cost (RAMC) metrics and specifies high-level design and sustainment metrics to track. DoDI 5000.91 requires using ADVANA data for visibility and management of performance-based sustainment metrics and costs. SCAIT supports the requirement by providing RAMC analyses of supply chain operation based on ADVANA data. The SCAIT RAMC analyses are actionable and suggest process improvements.

Air Force Policy requires adoption of CBM+ in weapon system sustainment. The policy goal is to improve availability and reduce life-cycle costs for weapon system through data-driven technologies, processes, and procedures that enable improved maintenance and logistics practices. Currently, data processing requires substantial effort of highly qualified analysts. Optimization of sustainment processes can improve availability of the Air Force aircraft. The earlier, smaller-scale deployment of SCAIT AI analytics as reporting service has provided demonstrable benefits to the 448th sustainment operation.

For each of the selected LRU part fleets, SCAIT can automatically

1. Pull ADVANA Data
2. Run AI
3. Report RAMC Anomalies (Monitoring)
4. Report RAMC Actionable Findings (Observability)

SCAIT analysis results can be accessed through Monitoring and Observability dashboards and reports.

- Monitoring detects anomalous KPIs. There are 25 KPIs monitored. The results are presented as

Executive dashboards and reports.

- Observability supports actionable findings and in-depth root cause analysis. There are twice as many detailed analysis results as the KPIs, about 50 at all. The results are presented as Engineering dashboards and reports.

This contract will expedite the operational use of SCAIT analytics in day-to-day sustainment processes at Tinker AFB, Hill AFB, and Robins AFB and support compliance with CBM+ Policy Objectives. The effort shall include Supply Chain processes performance improvement recommendations for selected repairable LRUs.

3.0 Scope

Two analytics tools have been developed within SCAIT; Reliability Digital Twin (RDT) analyzes maintenance processes, Reverse Logistics Digital Twin (RLDT) analyzes logistics processes. These two tools incorporated in SCAIT will generate AI Analytics Reports for each of selected LRUs each month. In these tasks, SCAIT analyses shall be based on data extracted by the Contractor from ADVANA/BLADE data repositories for the LRUs specified by the Government. The goal of project is scaling up SCAIT analyses across sustainment processes at AFSC 448th SCMW.

The scope of this PWS includes deploying SCAIT Software as a Service (SaaS) App in a Pilot Project. The following SaaS services shall be provided through the project period and will be consumed by Air Force users as a web App, included into the SaaS monthly subscription.

C SCAIT SaaS App deployment on DoD P1 private cloud with ADVANA data source. This includes achieving and sustaining continuous Authority To Operate (cATO), and Document Management for AI reports.

A SCAIT AI setup and support for selected LRU parts. Data for LRUs selected across the 448th SCMW sustainment operations at Tinker AFB, Hill AFB, and Robbins AFB will be analyzed every month through the performance period of the proposed effort.

U SCAIT SaaS App on-line availability for selected AFSC users and user support

The Contractor shall provide engineering services to promote and quantify the business impact of the SCAIT deployment on the AFSC processes. The Contractor shall identify best practices for all SCAIT users and shall document and deliver its findings via quarterly reports. The following Engineering Service will address the above requirements:

B Business Case Analyses (BCA) and Business Impact Support for the operational use of SCAIT under the following categories.

1. Reliability and Maintainability for RCM
2. Combined RDT/RLDT analysis and user success stories
3. PBL Contracts: Baseline assessment, SLAs, and monitoring
4. Mission-Capable Rate, Materiel Availability
5. Broader AI integration
 - a. Downstream post-AI processing and fusion with other data
 - b. Secure integration with other DoD data sources

4.0 Requirement/Description of Services

Contractor shall support the 448th SCMW by making SCAIT analysis capability available as SaaS App. This includes results of applying RDT AI and RLDT AI tools to Air Force data from selected LRUs.

This PWS requires the following lines of contractor effort within the scope of providing the SCAIT SaaS App subscription services:

C: Deploy SCAIT SaaS App on DoD network under DoD PlatformOne (P1) continuous Authority To Operate (cATO) and achieve its continuous availability for selected Air Force users.

A: Scale SCAIT processes assessments and performance improvement recommendations to 100 selected reparable LRUs. Provide access to the SCAIT SaaS App for 15 designated Air Force users. Provide SCAIT analysis reports as a collection of stand-alone documents accessible on-line.

U: Provide user support and user training to the designated Air Force users.

The following CDRLs shall be provided during the tasks described in this section:

CDRL A001 AI Analytics Results

4.1 DoD Cloud Setup

Contractor shall deploy SCAIT cloud App on DoD network under PlatformOne (P1) continuous Authority To Operate (cATO) and achieve its continuous access by selected Air Force users.

- The initial setup and P1 deployment of SCAIT SaaS shall be completed at 3 months ACA
- The full setup, and standing up Tech Support service, shall be completed in 6 months ACA

4.2 Data Preparation

Initial setup and data source preparation for the SCAIT analyses (RDT AI and RLDT AI reports for each of the selected LRUs) shall be completed in the first six months of the project, at an average pace of setting up four LRU parts per week. The setup effort shall include data reconciliation, validation, and updates to data cleansing logic necessary to set up the month-over-month reporting. It also shall include configuration setup and tuning of RDT AI and RLDT AI algorithms.

Data Reconciliation effort shall cross-validate different data sources to make sure data quality is acceptable. Data validation and reconciliation checks shall include:

- a. Data Quality Checks implemented in RDT AI
- b. Data Quality Checks implemented in RLDT AI
- c. RLDT AI flows reconciliation with NRTS Data (D200 data for 7SC reports)
- d. Cross-validation of Retrogrades independently computed by RDT AI and RLDT AI
 - Retrogrades in RDT AI are tracked removals sent to Depot as computed in RDT AI
 - Retrogrades in RLDT AI is given by retrograde flow computed in RLDT AI

Data preparation effort shall review the data quality metrics including comparisons of the variables computed in the two ways and deem the match acceptable or not acceptable. If the data reconciliation or data validation fails because of Air Force data issues for initially selected LRU, Air Force will select other LRU with better data as a replacement.

4.3 RLDT AI Analytics

RLDT AI predicts and optimizes Supply-Demand Balance in supply chain processes. RLDT AI is centered on the Explainable AI model for predicting supply chain process performance. In addition to the model reporting, AI functions covered by the RLDT AI results shall include:

- a. Repair demand prediction based on fleet usage that also accounts for seasonality,
- b. Process performance predictions including logistics delays, repairs time in Depot operations, part stocks, and part flows driven by the demand,

- c. Prediction for part availability risk using analysis approaches from finance industry,
- d. Quantifying alternatives for reducing cost while controlling risk: reliability vs logistics delay vs Depot performance vs spare parts,
- e. CBM+ cost impact and requirements

4.4 RDT AI Analytics

RDT AI predicts and optimizes reliability and maintenance demand for an asset or a reparable part. RDT AI uses the Explainable AI model for reliability as a foundation for Reliability- Centered Maintenance (RCM) analysis. In addition to the model reporting, AI functions covered by the RDT AI results shall include:

- a. Statistical Process Control for Bad Actor parts and aircraft
- b. Subpopulation analysis detecting reliability change for selected group of parts
- c. Early detection of recent change in fleet reliability
- d. Reliability centered prediction of various demand components related to line maintenance, Depot maintenance, No Fault Found, and Bad Actors

4.5 Data Security

For SCAIT Pilot Project, most aspects for protection of Government data will be ensured by P1 policies for host cloud services and ADVANA policies for the data used as SCAIT source.

- For the AI reports generated by SCAIT, protection of Government data shall be the same as in earlier SBIR Ph 3 projects.
- Data ownership, licensing, delivery, and disposition instructions specific to the relevant types of Government data and Government-related data shall be the same as in earlier SBIR Ph 3 projects.
- Data Ownership. For SCAIT app, data rights are the same as in earlier SBIR Ph 3 projects
- Data Breach. For SCAIT app, security, including addressing potential data breaches, is ensured through P1 deployment via P1 cATO
- Data Jurisdiction. For SCAIT app, data jurisdiction is ensured through P1 deployment
- To ensure compliance with regulations for Government records management policies, for SCAIT app, input data sets come from ADVANA as a system of record. The AI reports generated will be stored in P1 and can be transferred into Government Records Management system as requested.

4.6 Service Availability

For SCAIT Pilot Project, Service Level Objectives (SLO) will be pursued for most cloud services. The performance metrics in these SLOs will be measured and reported to serve as basis for the SLA in the follow-on SCAIT deployment project, see section 5.2. SLOs for availability of LRU AI Analysis Reports will include:

- a. Each new LRU part setup ready one week after LRU data IAW Section 4.2 is provided
- b. For SCAIT, AI runs for already set up LRU parts repeated monthly with current data.

Table below has additional SLO for SCAIT SaaS App on-line availability

Performance Requirements	Performance Threshold	Monitoring Method
APPLICATION AVAILABILITY		
Unscheduled application downtime	Customer meets application availability thresholds; Equal or fewer than 40 hours	monthly review of system metrics; section 5.2
Scheduled application downtime	Customer meets application availability thresholds; Equal or fewer than 100 hours	monthly review of system metrics; section 5.2

Availability of user access to the LRU AI Analysis Reports generated by SCAIT will be under Service Level Agreement (SLA). This will allow operational performance of AI analysis use in business processes as good as or better than in the earlier SBIR Ph 3 projects, see table below.

Performance Requirements	Performance Threshold	Monitoring Method
AI REPORTS AVAILABILITY		
Unscheduled reports downtime	Customer meets reports availability thresholds; Equal or fewer than 30 hours	monthly review of system metrics; section 5.2
Scheduled reports downtime	Customer meets reports availability thresholds; Equal or fewer than 80 hours	monthly review of system metrics; section 5.2

Service Availability levels for the SCAIT App achievable by Contractor are subject to availability and performance of government-provided services of P1. Redress for the SLA violation shall be provided in the form of credits for additional part analysis, beyond the part count limit in the license.

4.7 Technical Support

Technical Support of the user will be focused on Customer Success. In first 6 months of the project, while Pilot deployment of SaaS is yet partial and incomplete, the Tech Support will be provided via regular meetings of User Group consisting of the Air Force end users of SCAIT. In this period, Tech Support will be provided via email.

Full Help Desk Technical Support of users shall be provided after 6 months ACA. It shall be initiated via web service and have monitored Service Levels. Service Level Objectives (SLO) in table below will be pursued, measured, and reported for the Help Desk Tech Support.

Performance Requirements	Performance Threshold	Monitoring Method
HELP DESK SUPPORT		
Average speed to answer requests	80% answered <2 business days	monthly review of call handling activity reports; section 5.2
Average speed to resolution requests	65% resolved <10 business days	monthly review of call handling activity reports; section 5.2

Technical Support service levels for the SCAIT App achievable by Contractor are subject to availability and performance of government-provided services of P1.

4.8 Government Provided PlatformOne Services

Government shall provide contractor with required access to services of DoD PlatformOne (P1). This includes direct MIPR funding of P1 Billing for onboarding and hosting of the SCAIT SaaS App on P1 through the performance period.

The service levels for the SCAIT App achievable by Contractor are subject to availability and performance of government-provided services of P1

- Only response time of contractor can be specified as the service level. The DoD P1 response that might be required for the resolution is above and beyond that.
- The resolution of issues is subject to availability and performance of downstream DoD services for P1 and ADVANA including: data access, quality of the data in ADVANA, contractor personnel security management (e.g., providing access to new personnel), user access management, etc.
- Government-provided Tech Support for using ADVANA/BLADE for data access is required

4.9 Government Provided Information and Data Access

Government end-users/customers shall provide part definition information for each part analyzed by SCAIT. This information is unavailable in AF databases accessible by the contractor

1. WUC, NIIN, and MDS lists for each part
2. The lists must include historical data. For example, a NIIN might have been used several years ago, but is currently retired and not used. SCAIT pulls at historical data for its AI analysis and requires accurate historical part definition information.

Government shall provide Contractor personnel with supporting documents needed to get SCAIT App on P1 access to the required ADVANA/BLADE data, including but not limited to

1. Privacy Impact Assessment (PIA) DD2930 and PII Confidentiality Impact Level Template (PCIL) – no Personally Identifiable Information (PII) is used by SCAIT
2. Data Sharing Agreement (DSA) as required to access ADVANA/BLADE data

Government shall provide contractor personnel with security credentials for ADVANA/BLADE data access, such as an approved form DD2875, subject to appropriate security procedures.

As a minimum, access to the following data items will be provided by the Air Force to the Contractor for each of the selected LRUs via ADVANA/BLADE.

1. Operational Level Maintenance Data for all WUCs corresponding to selected LRU,
2. Configuration (Tracking) data for selected LRU,
3. Aircraft usage data for aircraft that host the selected LRU,
4. Optionally, data defining subpopulation of LRUs for subpopulation analysis.
5. Logistics transactions data for all NIINs corresponding to selected LRU,
6. MICAP event data for MICAPs involving the selected LRU,
7. Optionally, aircraft fleet usage forecast for future years, to improve demand forecasting.
8. Additional data required for validating integrity of the data pulls. For example, NRTS Data (D200 data for 7SC reports) is required by section 4.2.

5.0 Engineering Requirements/Services

The contractor shall provide engineering/technical services as described below in this section. The services include performance reporting, project management, documentation and data management, user training and system integration of AI Analysis results with existing Air Force business processes. This section of PWS requires Business Impact engineering services for promoting success of the Air Force Customer organization through the SCAIT AI capability deployment. The services shall include:

AI Analytics Results delivered, aggregated, and persistently available for all selected LRUs, to all users, as a collection of all generated SCAIT AI report documents.

Monthly Reporting of user issues, needs, and best practices. Contractor's SMEs shall work with AF end-users to help integrating SCAIT results into AFSC business processes and report the progress. This task shall support end users through User Group work meetings held at regular pace, e.g., biweekly, for first 6 months of this contract. After the 6 months, monthly review of Tech Support issues will be held and SLO/SLA metrics collected IAW PWS sections 4.6 and 4.7. This service task will also provide user documentation and user training.

Quarterly Reviews of Business Case Analyses (BCA) and Executive-level Success Stories for the operational use of SCAIT. This effort will be supported by Contractor's SMEs and subcontractors. The effort will be prioritized by AF PM in the performance period of the project, within contract scope. The effort will involve Advisory Group established with approval of Air Force PM and including AF SMEs.

In support of this project, the following tasks shall be performed:

5.1 AI Analytics Results

RLDT AI predicts and optimizes Supply-Demand Balance in supply chain processes. RLDT AI is centered on the Explainable AI model for predicting supply chain process performance. Reverse Logistics report data: RLDT AI results for selected LRUs

RDT AI predicts and optimizes reliability and maintenance demand for an asset or a repairable part. RDT AI uses the Explainable AI model for reliability as a foundation for Reliability- Centered Maintenance (RCM) analysis. Reliability report data: RDT AI results for selected LRUs

The following CDRLs shall be provided during the tasks described in section 5.1:
CDRL A001 Analytics Results

5.2 Monthly Summary Report

This task will have customer success focus. It will monitor progress of user support and delivery of AI Analytics Results to the users. The contractor shall submit a Summary Report on a monthly basis. This report shall contain at a minimum:

- Executive reporting of Summary Material for the completed analytics reporting work
- Cumulative Feedback
- Progress of the previous time period
- Issues/concerns and associated recommendations
- Projected activities for the next reporting period

The following CDRLs shall be provided during the tasks described in section 5.2; CDRL A002 Monthly Summary Report

5.3 Kick-off Meeting

The Contractor shall coordinate with the Government Technical Manager to schedule a Project Kickoff Meeting as soon as practicable after the project start. Kickoff Meeting Agenda shall include:

- a) Major milestones

- b) Roles and Responsibilities
- c) Risks, issues/concerns and associated recommendations
- d) Review of Action Items

The contractor shall provide presentation materials 3 business days before the meeting. Kick-off meeting location shall be presented either at Hill AFB, an immediate surrounding area (within 20 miles from Hill AFB), or by telecom.

The following CDRLs shall be provided during the tasks described in section 5.3;

- CDRL A003 Meeting Agenda
- CDRL A004 Meeting Minutes
- CDRL A005 Technical Report (*Action Items*)

5.4 Quarterly Program Status Reviews

This task will focus on monitoring and supporting progress of SCAIT AI adoption and efficiency of Supply Chain Management at Air Force 448 SCMW. The task will include Briefing and collecting feedback of Advisory Group approved by the Air Force Project Manager. The briefings shall include use cases, success stories, and user reports for SCAIT.

The contractor shall participate in program status reviews, as approved by the Air Force Project Manager. The reviews may be performed on site or by Conference calls. The purpose of these meetings shall be to brief the progress of each assigned task, and to solicit input and concurrence for planned workloads.

The contractor shall generate an informal agenda and deliver it to Air Force prior to each meeting.

The contractor shall be responsible for tracking Action Items.

The first Quarterly Program Status Review shall be held at the 90 days milestone marking start of regular monthly delivery of the reports.

The following CDRLs shall be provided during the tasks described in section 5.4:

- CDRL A003 Meeting Agenda
- CDRL A004 Meeting Minutes
- CDRL A005 Technical Report (*Action Items*)

5.5 Final Summary Report

The contractor shall provide a Final Summary Report. The final summary report shall include the summation of the work accomplished and prospective benefit to be realized from the technology. The report shall not contain any classified or proprietary information and written in such a way that the report may be publicly released by the United States Government.

The following CDRLs shall be provided during the tasks described in section 5.5;

- CDRL A006 Final Summary Report

6.0 Period of Performance

The period of performance for this task shall be for 15 months following award.

7.0 Deliverables

7.1 Data Deliverables

CDRL Listing

PWS Para.	Description	CDRL/DID Number	Freq.	As of Date	Date of 1 st Submission	Date of Subsequent Submission
4.0 5.1	AI Analytics Results	CDRL A001/ DI-MISC-80508B/T	As Req.	N/A	90 days ACA	N/A
5.2	Monthly Summary Report	CDRL A002/ DI-MGMT-81928/T	Monthly	N/A	90 days ACA	N/A
5.3 5.4	Meeting Agenda for Program Status Review/Technical Interchange Meetings	CDRL A003/ DI-ADMN-81249C	Quarterly	N/A	Two weeks before scheduled meetings.	15 th of the month after each Quarter
5.3 5.4	Meeting Minutes for Program Status Review/Technical Interchange Meetings	CDRL A004/ DI-ADMN-81250C	1 Time	N/A	Two weeks after scheduled meetings.	ASREQ
5.3 5.4	Technical Report	CDRL A005/ DI-MISC-80508B/T	As Req.	N/A	N/A	N/A
5.5	Final Summary Report	CDRL A006/ DI-MISC-80711A	1 Time	N/A	425 DAC	N/A

Government Inspection and Acceptance of Deliverables

Documentation deliverables shall be delivered IAW this PWS, unless otherwise specified by the Administrative Contracting Officer (ACO) through a contract modification or letter. The Government has the inherent right to disapprove any deliverable, whether data or non-data. Therefore, the TPOC will have the right to reject or require correction of any deficiencies found in deliverables. In the event of rejection of any deliverable, the TPOC, with coordination with the ACO, will notify the Contractor in writing of the specific reasons why the deliverable was rejected. The Contractor shall have 15 calendar days or unless designated otherwise through an ACO letter, to correct the rejected deliverable and resubmit to the TPOC for re-inspection. The Government must approve final versions of data deliverables.

8.0 Government Technical Contacts

TPOC: 748 SCMG/EN
6071 Gum Lane, Bldg 1223
Hill Air Force Base, Utah 84056-5826